# SECTION 9 – CRYOSTABLE MAGNET RAMPING PROCEDURE

**DANGER!!**

**MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!**

**REVIEW AND FULLY UNDERSTAND THE SUPERCONDUCTING MAGNET SAFETY REQUIREMENTS SECTION ( INTRODUCTION, SECTION 5–3 ) OF THIS MANUAL.**

**FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN THE SAFETY CHECKLIST SECTION ( INTRODUCTION, SECTION 5–5, TABLE 5–1 ) OF THIS MANUAL.**

**Description:**

The Cryostable magnet Ramping current is approximately 745 amps for 1.0T and 742 amps for 1.5T. This procedure utilizes Voltage Control for ramping. If Ramp and Shim Cables were not ordered with the magnet, Ramp Cable Kit P/N 2135435 will be needed to Ramp the magnet and Shim Cable Kit P/N 2135558 will be needed to shim the magnet.

**SECTION 9 – CRYOSTABLE MAGNET RAMPING PROCEDURE ( continued )**

**WARNING!**

**THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:**

**MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5–3 OF THE INTRODUCTION ( CRYOGEN SAFETY ).**

**NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.**

**POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE INTRODUCTION, SECTION 5.**

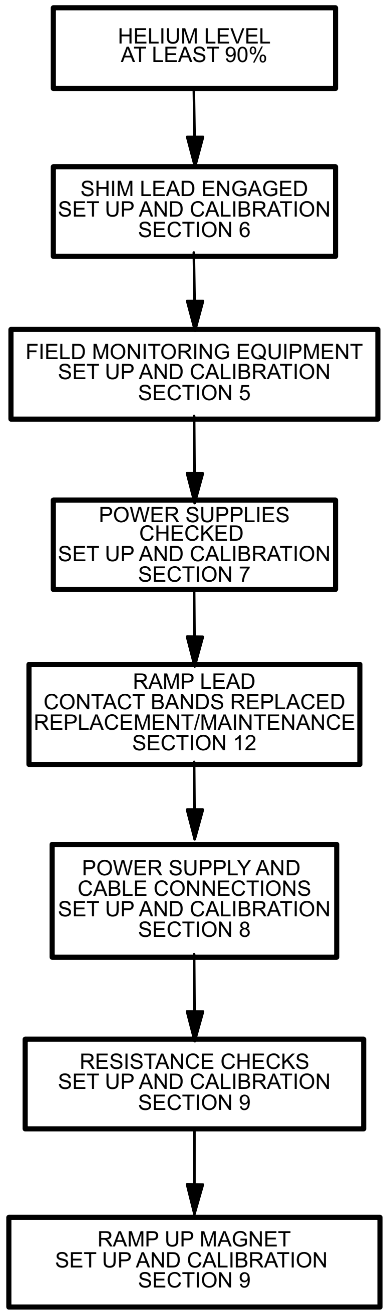
**POST “RAMPED MAGNET” AND “AUTHORIZED PERSONNEL ONLY” WARNING SIGNS AT THE MAGNET ROOM ENTRANCE AND AT THE MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED.**

**REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**

**MAKE SURE THE MAGNET RUNDOWN UNIT IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET UP AND CALIBRATION, SECTION 1–5.**

**MAKE SURE THE MAGNET IS AT LEAST 90% FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.**

## SECTION 9 – CRYOSTABLE MAGNET RAMPING PROCEDURE ( continued )




**RAMP FLOWCHART**
ILLUSTRATION 9–1

## 9–1 PREPARATION

1. Set up the field monitoring equipment, Teslameter and Teslameter Probe, in conformance with SET UP AND CALIBRATION, Section 5.
2. Perform Magnet Electrical Checks in conformance with FUNCTIONAL CHECKS, Section 3.
3. Make sure the magnet is at least 90% full of helium before Ramping the magnet.

**WARNING!**

**THE MAGNET CAN BE QUENCHED IF THE MAGNET POWER SUPPLY EXPERIENCES LARGE OUTPUT VOLTAGE FLUCTUATIONS AND/OR EXCESSIVE RIPPLE. MAKE SURE THE POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.**

4. Make sure that the Main Power Supply is installed, checked and adjusted in conformance with the Vendor Manual supplied with the unit.
5. Make sure the Shim Lead Assembly is “Engaged” in conformance with SET UP AND CALIBRATION, Section 6.
6. Make sure that Input Power Cable to the Power Supply is disconnected.
7. Connect the Power Supply to the magnet by making all cable connections in conformance with SET UP AND CALIBRATION, Section 8 ( Electrical Connections For Ramping And Shimming ).
8. Record the Main Coil connection polarity in DATA SHEETS, Table 6–1.
9. Set all power supply heater switches to the OFF position. Set CURRENT ADJUST and VOLTAGE controls to 0 (full CCW).
10. Connect the Input Power Cable to the Main Power Supply.
11. Make sure He Vent Valve ( V2 ) is closed.

## 9–2 RESISTANCE CHECKS

1. Check Switch Heater and Shim Coil resistances in conformance with FUNCTIONAL CHECKS, Section 3.
2. Make sure CURRENT ADJUST and VOLTAGE controls on the Main Power Supply are off ( full CCW ).

**Note**

All “Main Coil Driving Voltages” provided in this procedure will be equal in magnitude but opposite in polarity for “Reverse Ramped” magnets.

## 9–2 RESISTANCE CHECKS (continued)

**WARNING!**

**MAKE SURE MAIN HEATER SWITCH IS OFF DURING THE RESISTANCE CHECKS.**

3. Set the MAIN POWER and POWER ON switches to ON ( Switches located on both the front and back of the Main Power Supply ).
4. Set HEATER 2 SHIM AXIAL switch to 1 ( on ) and observe current rise in ammeter ( 800–820 mA ) to verify circuit continuity. Make sure Main Heater Switch is off.
5. Connect a Digital Voltmeter ( DVM ) to the end of the Voltage Sense Leads.
6. Set CURRENT ADJUST COARSE control on power supply to maximum ( full CW ).
7. Observe the Main Power Supply Ammeter and slowly turn the VOLTAGE control ( CW ) to set 750A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
8. Record the voltage reading on the ( DVM ) in the DATA SHEET tab, Table 6–1.

**WARNING!**

**A VOLTAGE READING GREATER THAN 150 MILLIVOLTS AT 750 AMPS INDICATES UNACCEPTABLE INTERNAL CONTACT RESISTANCE OF THE LEAD EXTENSIONS. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET INCREASING BOILOFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING.**

9. Perform one or more of the bulleted steps below, as necessary, if the DVM voltage is greater than 150mV.
   - Wait approximately 1 minute with current running; readings may drop as Power Lead Extensions cool.
   - Tighten the nuts on top of the Hold Down Tool.
   - If the reading still exceeds 150 mV, turn VOLTAGE and CURRENT ADJUST controls to zero ( full CCW ), turn off Magnet Power Supply input power, then check / tighten bolts securing the Ramp Cables to the Power Supply and Ramp Leads Extensions. Lift and reseat the Ramp Leads. Repeat Steps 6 through 9.
10. Set the power supply VOLTMETER SELECT SWITCH to MAIN POWER SUPPLY position. ( This will display the output of the power supply monitored at the output lugs. )

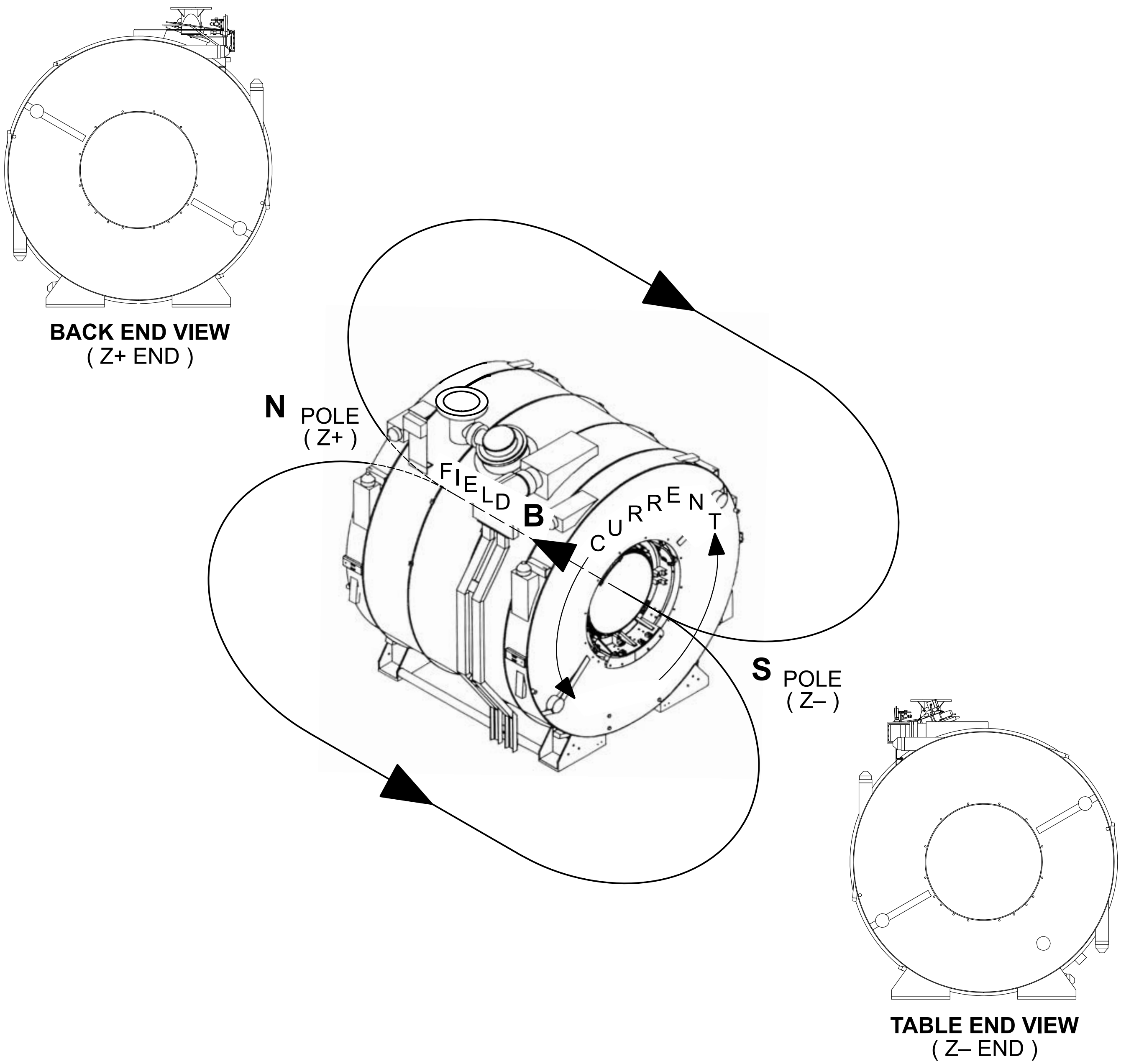
**Note**

A voltage less than 2.2V at 750A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, follow the procedures in Step 9 for adjusting contact resistance.

## 9–2 RESISTANCE CHECKS (continued)

11. Gradually increase the VOLTAGE ADJUST control to pass 750A through the Main Power leads, Lead Extensions and persistent Main Switch while observing the Power Supply Voltmeter. If the voltage exceeds 2.2V, then check/tighten the bolts securing the Ramp Cables to the Power Supply and Ramp Lead Extensions.

12. Turn the CURRENT ADJUST and VOLTAGE controls off ( full CCW ) and continue with the ramping procedure after completion of Step 11.




**MAGNET POLARITY RAMPED NORMAL**
ILLUSTRATION 9–2

## 9–3 RAMPING FOR 1.0T

**CAUTION**

**If a Quench occurs during ramping, immediately turn VOLTAGE control and CURRENT control of the Main Coil Power Supply to zero (fully CCW). A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the Vent System.**

**WARNING!**

**MAKE SURE THAT THE AXIAL SHIM SWITCH HEATERS ARE ON DURING RAMPING TO PREVENT COIL DAMAGE AND MAGNET QUENCH. GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.**

**IF MAGNET IS A 1.0T MOBILE, TRANSVERSE 1 AND 2 SWITCH HEATERS MUST BE “ON”. THE SHIM POWER SUPPLY IS REQUIRED FOR TRANSVERSE HEATERS.**

**Note**

Ice will form around the Ramp Lead Hold Down Tool Flow Holes during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure valve V2 is closed.

2. Make sure VOLTAGE ADJUST AND CURRENTS ADJUST controls are at zero (full CCW).

**Note**

The Axial Switch Heater current is supplied by the Main Service Ramp Supply. The Transverse 1 and Transverse 2 Switch Heaters currents are supplied by the Shim Supply. Do not use the Axial Heater on the Shim Supply as this will increase boil off.

3. Set HEATER 2 SHIM AXIAL Switch, on the Main Power Supply, to 1 ( ON ).

4. Set Main Ramp Power Supply MAIN POWER Switch to ON.

5. Adjust the CURRENT ADJUST controls to maximum ( full clockwise ). Adjust the VOLTAGE ADJUST control to minimum ( full counterclockwise ).

6. Set HEATER 1 MAIN switch to 1 ( on ). Wait 3 minutes.

7. Set the power supply VOLTMETER SELECT SWITCH to MAIN COIL position.

## 9–3 RAMPING FOR 1.0T ( continued )

8. Access spreadsheet on floppy disk sent with magnet ATR. Obtain currents from spreadsheet as indicated below.

   Cx10_15.XLW
   1.0T PARKING
   1.5T PARKING

   INPUT
   AXIAL 2, 4, 6 CURRENTS
   DECOMP 2, 0 4, 0 6, 0 PPM VALUES

   With no shim power connected during ramp, park magnet at frequency indicated by the spreadsheet.

TABLE 9–1
**RAMPING VOLTAGE VERSUS CURRENT**

| MAIN COIL DRIVING VOLTAGE | MAIN COIL CURRENT |
|---|---|
| 5.00 VOLTS | FROM 0 TO 100 AMPS |
| 4.6 VOLTS | FROM 100 TO 200 AMPS |
| 4.2 VOLTS | FROM 200 TO 300 AMPS |
| 3.8 VOLTS | FROM 300 TO 400 AMPS |
| 3.3 VOLTS | FROM 400 TO 500 AMPS |
| 2.5 VOLTS | FROM 500 TO 550 AMPS |
| 2.0 VOLTS | FROM 550 TO 600 AMPS |
| 1.5 VOLTS | FROM 600 TO 625 AMPS |
| 1.0 VOLTS | FROM 625 TO 650 AMPS |
| 0.75 VOLTS | FROM 650 TO 675 AMPS |
| 0.60 VOLTS | FROM 675 TO 700 AMPS |
| 0.50 VOLTS | FROM 700 TO 720 AMPS |
| 0.30 VOLTS | FROM 720 TO 730 AMPS |
| SLOWLY LOWER DRIVE VOLTAGE FROM 0.3 VOLTS | 730 AMPS |
| 0.1 VOLTS | WHEN MAGNET REACHES 0.990 T |
| 0.025 VOLTS | WHEN MAGNET REACHES 0.990 T |
| EQUALIZE | WHEN MAGNET REACHES 1.001 T |

9. Turn power supply VOLTAGE ADJUST control until a reading of 5.00 volts is observed on the power supply digital VOLTMETER.

**Note**

Measured inductance should be approximately 5.0 henrys. If the calculated value is between 4.8 – 5.2 Henries continue with the procedure. If the calculated value is outside this range, discontinue ramping and measure Main Coil Resistance ( See FUNCTIONAL CHECKS, Section 3 ). Contact the Region MAC Team Representative.

**9–3 RAMPING FOR 1.0T ( continued )**

10. Estimate the system inductance by measuring current change over a 10 second ramping interval:

L ( inductance ) = 10 x Voltage / Current Change

**Note**

This method will give inaccurate values of inductance when the current is less than 200 Amps.

**Note**

The Teslameter will lock on when magnet current rises to approximately 515 amps ( using a Range 5 Probe ). This usually occurs between 0.6 and 0.7 Tesla (25.545900 MHz – 29.803550 MHz).

**WARNING!**

**THE 1.0T MAGNET CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. WHEN ADJUSTING CURRENT IN STEP 10 BELOW, TURN CURRENT CONTROL IN A SLOW, CONTINUOUS MOTION. BE CAREFUL NOT TO JERK THE CURRENT CONTROL.**

**Note**

Allow Main Coil Driving voltage to decay between each of the steps below ( e.g. in Step 11 adjust the Main Coil Driving voltage to 4.6V and do not readjust until Step 12 ).

11. When the magnet current reaches 100 amps, slowly adjust the VOLTAGE ADJUST control, counterclockwise, for 4.6 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter ( Voltmeter Select Switch in the Main Coil position ).

12. When the magnet current reaches 200 amps, slowly adjust the VOLTAGE ADJUST control for 4.2 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

13. When the magnet current reaches 300 amps, slowly adjust the VOLTAGE ADJUST control for 3.8 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

14. When the magnet current reaches 400 amps, slowly adjust the VOLTAGE ADJUST control for 3.3 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

15. When the magnet current reaches 500 amps, slowly adjust the VOLTAGE ADJUST control for 2.5 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

16. When the magnet current reaches 550 amps, slowly adjust the VOLTAGE ADJUST control for 2.0 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

17. When the magnet current reaches 600 amps, slowly adjust the VOLTAGE ADJUST control for 1.5 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

18. When the magnet current reaches 625 amps, slowly adjust the VOLTAGE ADJUST control for 1.0 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

19. When the magnet current reaches 650 amps, slowly adjust the VOLTAGE ADJUST control for 0.75 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.

**9–3 RAMPING FOR 1.0T ( continued )**

20. When the magnet current reaches 675 amps, slowly adjust the VOLTAGE ADJUST control for 0.6 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
21. When the magnet current reaches 700 amps, slowly adjust the VOLTAGE ADJUST control for 0.5 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
22. When the magnet current reaches 720 amps, slowly adjust the VOLTAGE ADJUST control for 0.3 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
23. When the magnet current reaches 730 amps, slowly lower the VOLTAGE ADJUST control from 0.3 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
24. When the magnet current reaches 0.990 Tesla, slowly adjust the VOLTAGE ADJUST control for 0.1 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
25. When the magnet current reaches 0.999 Tesla, slowly adjust the VOLTAGE ADJUST control for 0.025 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
26. Check Teslameter and slowly adjust the VOLTAGE ADJUST controls, as required, to bring Magnetic Field to frequency shown on ATR. The total current will be approximately 745 amps. Allow final field to stabilize. The last two digits on the Teslameter should be the only digits changing.
27. Maintain field at final setting for 5 minutes before proceeding to Step 28.

**Note**

Observe voltage ( read on Power Supply Voltmeter with toggle switch in MAIN COIL position ). When the switch goes “persistent” the voltage across the magnet terminals will stabilize at 0.00.

28. Turn off Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go “persistent”.
29. Record current, frequency and lead extension voltage values at which the switch went “persistent” in DATA SHEETS, Table 6–1.

**WARNING!**

**MAKE SURE THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 6–1. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING OF THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.**

**Note**

Check that Teslameter does not decrease as the VOLTAGE control knob is turned to Zero. Only the last two digits on the Teslameter should change. If the field decreases as the VOLTAGE control knob is turned, the main coil switch is not persistent and the VOLTAGE control must be slowly adjusted to return to Parking Field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

30. When the switch goes “persistent”, **slowly** turn the power supply VOLTAGE control to zero over a two minute period ( Full CCW ).

**9–3 RAMPING FOR 1.0T ( continued )**

31. Gradually turn the CURRENT ADJUST control to zero ( over a one minute period ).

32. Set HEATER 2 SHIM AXIAL switch to 0 ( off ).

33. Set MAIN POWER AND POWER ON switches to OFF, on Main Power Supply and disconnect Input Power Cable.

34. Remove ramp lead hold down tool and replace bolts on lead assembly mounting plate.

**CAUTION**

**Step 35 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.**

35. After parking the magnet and disconnecting main power supply, plug and remove one ramp extension at a time in the following sequence.
    a. Open valve ( V2 ) to de–pressurize the cryostat to 0.25 psig. Close V2.
    b. Install a screw into the flow hole of only one of the ramp lead extensions. See Illustration 9–3.
    c. Remove all ice around the ramp lead port compression nut on the ramp lead extension that is being removed ( i.e. the ramp lead extension that has the flow hole plugged in Step b ). See Illustration 9–3.
    d. Unscrew the ramp lead port compression nut and remove the ramp lead extension from the magnet. Immediately replace the cap onto the ramp lead port.
    e. Repeat Steps b through e for the other ramp lead extension.

**Note**
Transverse coil currents will be removed in SET UP AND CALIBRATION, Section 10–2.

36. There may be small currents induced onto the Transverse Coils by the Ramping process. Make sure these currents are removed in conformance with SET UP AND CALIBRATION, Section 10–2 before Magnetic Probe Centering and Shimming the magnet.

**CAUTION**

**Cryostat exhaust flow rates and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve ( V2 ).**

**Note**
Read all flow rates from the bottom of the float ( ball ) on the flow meters.

**Note**
Flow rates may be temporarily elevated after ramping. Do not adjust them until after the magnet has had time to stabilize ( at least one day ).

**9–3 RAMPING FOR 1.0T ( continued )**

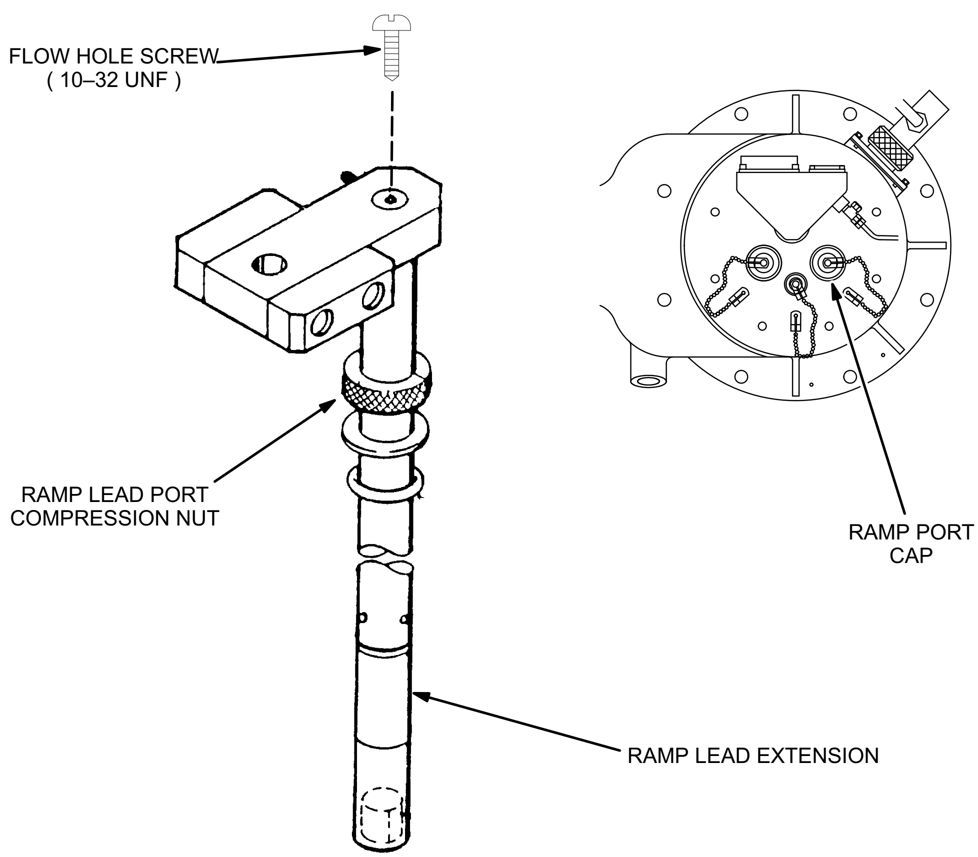
37. Make sure the following conditions are maintained. Re–check settings in three days and again after one week:

INSTRUMENTATION LEAD FLOWMETER ( F2 ) = 0.8 – 1.2 SCFH
SHIM LEAD FLOWMETER ( F1 ) = 1.8 – 2.2 SCFH
CRYOSTAT GAUGE PRESSURE = 0.25 – 0.50 psig

**Note**

If flow rate through F2 is less than 0.8 SCFH or the pressure gauge reads less than 0.25 psig, pressurize the vessel and "bubble test" all exhaust plumbing joints, relief valve and Shim Lead Connector. Make sure V2 is fully closed. Repair any leaks.

38. Proceed to SET UP AND CALIBRATION, Section 10 ( Shimming Preparation/Field Stabilization" ).




**RAMP LEAD EXTENSION AND RAMP PORT COMPRESSION NUT**
ILLUSTRATION 9–3

## 9–4 RAMPING FOR 1.5T

**CAUTION**

**If a Quench occurs during ramping, immediately turn VOLTAGE control and CURRENT control of the Main Coil Power Supply to zero ( fully CCW ). A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the Vent System.**

**WARNING!**

**MAKE SURE THAT THE AXIAL SHIM SWITCH HEATERS ARE ON DURING RAMPING TO PREVENT COIL DAMAGE AND MAGNET QUENCH. GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE "OFF" POSITION.**

**IF MAGNET IS A 1.5T MOBILE, TRANSVERSE 1 AND 2 SWITCH HEATERS MUST BE "ON". THE SHIM POWER SUPPLY IS REQUIRED FOR TRANSVERSE HEATERS.**

**Note**

Ice will form around the Ramp Lead Hold Down Tool Flow Holes during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure valve V2 is closed.
2. Make sure VOLTAGE ADJUST AND CURRENTS ADJUST controls are at zero ( full CCW ).

**Note**

The Axial Switch Heater current is supplied by the Main Service Ramp Supply. The Transverse 1 and Transverse 2 Switch Heaters currents are supplied by the Shim Supply. Do not use the Axial Heater on the Shim Supply as this will increase boil off.

3. Set HEATER 2 SHIM AXIAL Switch, on the Main Power Supply, to 1 ( ON ).
4. Set Main Ramp Power Supply MAIN POWER Switch to ON.
5. Adjust the CURRENT ADJUST controls to maximum ( full clockwise ).
6. Adjust the VOLTAGE ADJUST control to minimum ( full counterclockwise ).
7. Set HEATER 1 MAIN switch to 1 ( on ). Wait 3 minutes.
8. Set the power supply VOLTMETER SELECT SWITCH to MAIN COIL position.

## 9–4 RAMPING FOR 1.5T ( continued )

9. Access spreadsheet on floppy disk sent with magnet ATR. Obtain currents from spreadsheet as indicated below.

   Cx10_15.XLW
   1.0T PARKING
   1.5T PARKING

   INPUT
   AXIAL 2, 4, 6 CURRENTS
   DECOMP 2, 0 4, 0 6, 0 PPM VALUES

   With no shim power connected during ramp, park magnet at frequency indicated by the spreadsheet.

TABLE 9–2
**RAMPING VOLTAGE VERSUS CURRENT**

| MAIN COIL DRIVING VOLTAGE | MAIN COIL CURRENT |
|---|---|
| 6.0 VOLTS | FROM 0 TO 100 AMPS |
| 5.6 VOLTS | FROM 100 TO 200 AMPS |
| 5.2 VOLTS | FROM 200 TO 300 AMPS |
| 4.7 VOLTS | FROM 300 TO 400 AMPS |
| 4.3 VOLTS | FROM 400 TO 500 AMPS |
| 3.5 VOLTS | FROM 500 TO 550 AMPS |
| 3.0 VOLTS | FROM 550 TO 600 AMPS |
| 2.35 VOLTS | FROM 600 TO 625 AMPS |
| 2.0 VOLTS | FROM 625 TO 650 AMPS |
| 1.0 VOLTS | FROM 650 TO 675 AMPS |
| 0.75 VOLTS | FROM 675 TO 700 AMPS |
| 0.5 VOLTS | FROM 700 TO 720 AMPS |
| 0.3 VOLTS | FROM 720 TO 730 AMPS |
| SLOWLY LOWER DRIVE VOLTAGE FROM 0.3 VOLTS | 730 AMPS |
| 0.1 VOLTS | WHEN MAGNET REACHES 1.490 T |
| 0.025 VOLTS | WHEN MAGNET REACHES 1.499 T |
| EQUALIZE | WHEN MAGNET REACHES 1.501 T |

10. Turn power supply VOLTAGE ADJUST control until a reading of 6.00 volts is observed on the power supply digital VOLTMETER.

**Note**
Measured inductance should be approximately 10.3 henrys. If the calculated value is between 10.0 – 10.5 Henries continue with the procedure. If the calculated value is outside this range, discontinue ramping and measure Main Coil Resistance ( See FUNCTIONAL CHECKS, Section 3 ). Contact the Region MAC Team Representative.

**9–4 RAMPING FOR 1.5T ( continued )**

11. Estimate the system inductance by measuring current change over a 10 second ramping interval:

    L ( inductance ) = 10 x Voltage / Current Change

**Note**
This method will give inaccurate values of inductance when the current is less than 200 Amps.

**Note**
The Teslameter will lock on when magnet current rises to approximately 515 amps ( using a Range 5 Probe ). This usually occurs between 0.6 and 0.7 Tesla ( 25.545900 MHz – 29.803550 MHz ).

**WARNING!**

**THE 1.5T MAGNET CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. WHEN ADJUSTING THE CURRENT IN STEP 11 BELOW, TURN THE CURRENT CONTROL IN A SLOW, CONTINUOUS MOTION. BE CAREFUL NOT TO JERK THE CURRENT CONTROL.**

**Note**
Allow Main Coil Driving voltage to decay between each of the steps below ( e.g. in Step 12 adjust the Main Coil Driving voltage to 5.6V and do not readjust until Step 13 ).

12. When the magnet current reaches 100 amps, slowly adjust the VOLTAGE ADJUST control, counterclockwise, for 5.6 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter ( Voltmeter Select Switch in the Main Coil position ).
13. When the magnet current reaches 200 amps, slowly adjust the VOLTAGE ADJUST control for 5.2 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.
14. When the magnet current reaches 300 amps, slowly adjust the VOLTAGE ADJUST control for 4.7 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.
15. When the magnet current reaches 400 amps, slowly adjust the VOLTAGE ADJUST control for 4.3 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.
16. When the magnet current reaches 500 amps, slowly adjust the VOLTAGE ADJUST control for 3.5 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.
17. When the magnet current reaches 550 amps, slowly adjust the VOLTAGE ADJUST control for 3.0 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.
18. When the magnet current reaches 600 amps, slowly adjust the VOLTAGE ADJUST control for 2.35 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.
19. When the magnet current reaches 625 amps, slowly adjust the VOLTAGE ADJUST control for 2.0 Volts Main Coil Driving Voltage as indicated on the Power Supply Voltmeter.

## 9–4 RAMPING FOR 1.5T ( continued )

20. When the magnet current reaches 650 amps, slowly adjust the VOLTAGE ADJUST control for 1.0 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
21. When the magnet current reaches 675 amps, slowly adjust the VOLTAGE ADJUST control for 0.75 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
22. When the magnet current reaches 700 amps, slowly adjust the VOLTAGE ADJUST control for 0.5 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
23. When the magnet current reaches 720 amps, slowly adjust the VOLTAGE ADJUST control for 0.3 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
24. When the magnet current reaches 730 amps, slowly adjust the VOLTAGE ADJUST control for 0.3 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
25. When the magnet current reaches 1.490 Tesla, slowly adjust the VOLTAGE ADJUST control for 0.1 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
26. When the magnet current reaches 1.499 Tesla, slowly adjust the VOLTAGE ADJUST control for 0.025 Volts MAIN Coil Driving Voltage as indicated on the Power Supply Voltmeter.
27. Check Teslameter and **slowly** adjust the VOLTAGE ADJUST controls, as required, to bring Magnetic Field to frequency shown on ATR. The total current will be approximately 742 amps. Allow final field to stabilize. The last two digits on the Teslameter should be the only digits changing.
28. Maintain field at final setting for 5 minutes before proceeding to Step 29.

**Note**

Observe voltage ( read on Power Supply Voltmeter with toggle switch in MAIN COIL position ). When the switch goes “persistent” the voltage across the magnet terminals will stabilize at 0.00.

29. Turn off Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go “persistent”.
30. Record current, frequency and lead extension voltage values at which the switch went “persistent” in DATA SHEETS, Table 6–1.

**WARNING!**

**MAKE SURE THAT THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 6–1. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING OF THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.**

**Note**

Check that Teslameter does not decrease as the VOLTAGE control knob is turned to Zero. Only the last two digits on the Teslameter should change. If the field decreases as the VOLTAGE control knob is turned, the main coil switch is not persistent and the VOLTAGE control must be slowly adjusted to return to Parking Field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

### 9–4 RAMPING FOR 1.5T ( continued )

31. When the switch goes “persistent”, **slowly** turn the power supply VOLTAGE control to zero over a two minute period ( Full CCW ).

32. Gradually turn the CURRENT ADJUST control to zero ( over a one minute period ).

33. Set HEATER 2 SHIM AXIAL switch to 0 ( off ).

34. Set MAIN POWER AND POWER ON switches to OFF, on Main Power Supply and disconnect Input Power Cable.

35. Remove ramp lead hold down tool and replace bolts on lead assembly mounting plate.

**CAUTION**

**Step 36 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.**

36. After parking the magnet and disconnecting main power supply, plug and remove one ramp extension at a time in the following sequence.
    a Open valve ( V2 ) to de–pressurize the cryostat to 0.25 psig. Close V2.
    b Install a screw into the flow hole of only one of the ramp lead extensions. See Illustration 9–4.
    c Remove all ice around the ramp lead port compression nut on the ramp lead extension that is being removed ( i.e. the ramp lead extension that has the flow hole plugged in Step b ). See Illustration 9–4.
    d Unscrew the ramp lead port compression nut and remove the ramp lead extension from the magnet. Immediately replace the cap onto the ramp lead port.
    e Repeat Steps b through e for the other ramp lead extension.

**Note**
Transverse coil currents will be removed in SET UP AND CALIBRATION, Section 10–2.

37. There may be small currents induced onto the Transverse Coils by the Ramping process. Make sure these currents are removed in conformance with SET UP AND CALIBRATION, Section 10–2 before Magnetic Probe Centering and Shimming the magnet.

**CAUTION**

**Cryostat exhaust flow rates and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve ( V2 ).**

**Note**
Read all flow rates from the bottom of the float ( ball ) on the flow meters.

## 9–4 RAMPING FOR 1.5T ( continued )

**Note**

Flow rates may be temporarily elevated after ramping. Do not adjust them until after the magnet has had time to stabilize ( at least one day ).

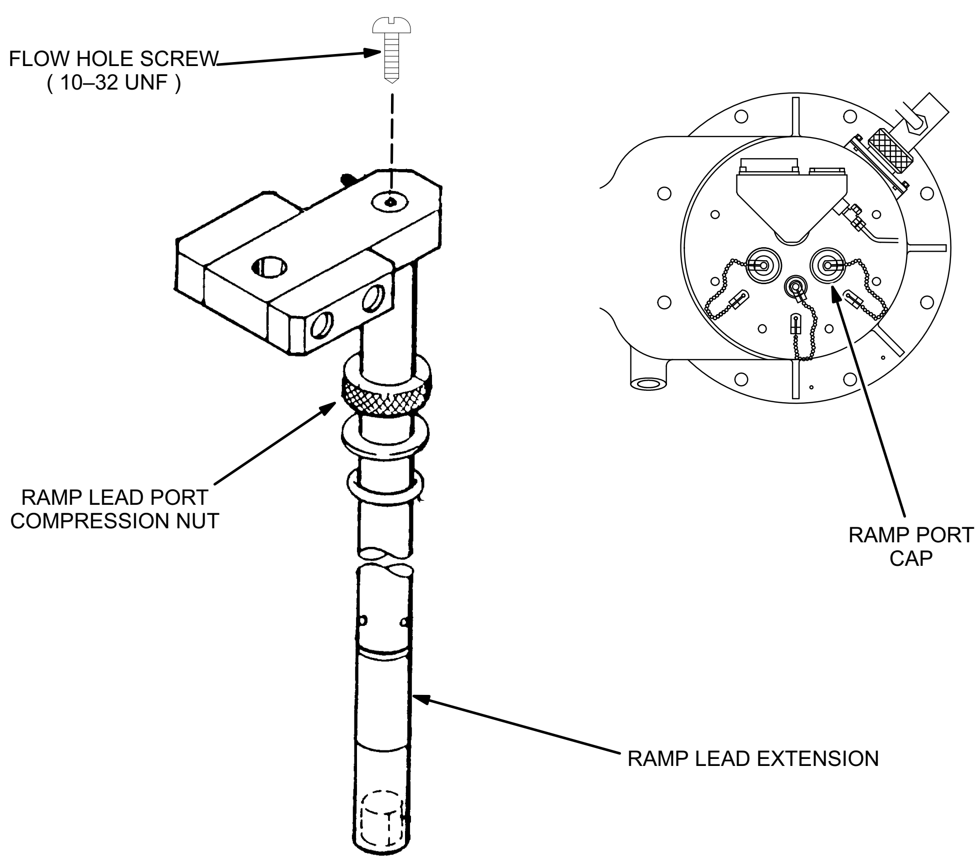
38. Make sure the following conditions are maintained. Re–check settings in three days and again after one week:

INSTRUMENTATION LEAD FLOWMETER ( F2 ) = 0.8 – 1.2 SCFH
SHIM LEAD FLOWMETER ( F1 ) = 1.8 – 2.2 SCFH
CRYOSTAT GAUGE PRESSURE = 0.25 – 0.50 psig

**Note**

If flow rate through F2 is less than 0.8 SCFH or the pressure gauge reads less than 0.25 psig, pressurize the vessel and "bubble test" all exhaust plumbing joints, relief valve and Shim Lead Connector. Make sure V2 is fully closed. Repair any leaks.

39. Proceed to SET UP AND CALIBRATION, Section 10 ( "Shimming Preparation/Field Stabilization" ).




**RAMP LEAD EXTENSION AND RAMP PORT COMPRESSION NUT**
ILLUSTRATION 9–4